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Advanced Machine Learning

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IMDB Movie Review Classification

The IMDB movie review dataset was used to build a model that predicts whether a movie review is positive or negative. Several different neural network designs were tested. The methods used changed the number of hidden layers, the number of hidden units, the loss function, the activation function, and the use of regularization techniques. The goal was to determine how these changes affected validation and test accuracy. The main measure used to compare the models was accuracy. Validation accuracy was used while developing and comparing the models, while test accuracy was used to measure performance on data that the models had not used during training.

The IMDB dataset contains 50,000 movie reviews. There are 25,000 reviews in the training dataset and 25,000 reviews in the test dataset. The model used the 10,000 most common words in the dataset. Each review was converted into a numerical format that could be used by the neural network.

The original 25,000 training reviews were divided into 15,000 training reviews and 10,000 validation reviews. The 25,000 test reviews were kept separate and were only used to evaluate the final models. This allowed the models to be trained on one group of reviews, compared using a separate validation group, and finally tested using data that had not been used during model development.

The baseline model used two hidden layers. Each hidden layer contained 16 units and used the Relu activation function. The output layer used a sigmoid activation function because the model was predicting one of two possible outcomes, positive or negative. The baseline model achieved a best validation accuracy of 89.17% at epoch 4. Its test accuracy was 86.25%. The baseline provided a reference point for the other models. Each change was compared against this model to determine how the change affected performance.

The first experiment changed the number of hidden layers. The one hidden layer model achieved a best validation accuracy of 88.93%. The three-hidden-layer model achieved 89.04% validation accuracy. The baseline with two hidden layers achieved 89.17%. This shows that adding another hidden layer did not automatically improve the model. The three-layer model was slightly below the baseline. The results suggest that increasing model complexity does not always lead to better performance.

The next experiment changed the number of units in the two hidden layers. The model with 32 units in each layer achieved 88.97% validation accuracy. The model with 64 units in each layer achieved 89.07%. These results were very close to the baseline. Increasing the number of units did not create a large improvement in validation accuracy. The 64-unit model had a test accuracy of 86.86%, compared with 86.25% for the baseline.

The MSE or mean squared error model achieved a validation accuracy of 88.73% and a test accuracy of 87.69%. The validation accuracy was lower than the baseline, but the test accuracy was higher. This shows that a model can have slightly lower validation accuracy while still performing well on the final test data. The tanh model achieved a best validation accuracy of 88.88% and a test accuracy of 86.20%. The results were slightly below the baseline. In this experiment, changing the activation function from Relu to tanh did not improve the model's

performance. Regularization techniques were tested to see whether they could reduce overfitting and improve performance on unseen data.

Dropout

- The model using 20% dropout achieved 88.80% validation accuracy and 87.11% test accuracy.
- The model using 50% dropout also achieved 88.80% validation accuracy. However, its test accuracy was 88.09%, which was the highest test accuracy among the models tested.

The dropout models also showed a different training pattern. Training accuracy increased more slowly than in the baseline model. This is expected because dropout prevents some units from being used during training.

L2 Regularization

- The L2 regularization model achieved 88.88% validation accuracy and 86.59% test accuracy.
- The validation accuracy was close to the other models, but the test accuracy was below the baseline.

Model	Hidden Layers	Hidden Units	Best Validation Accuracy	Test Accuracy
One Hidden Layer	1	16	88.93%	N/A

Model	Hidden Layers	Hidden Units	Best Validation Accuracy	Test Accuracy
Baseline	2	16, 16	89.17%	86.25%
Three Hidden Layers	3	16, 16, 16	89.04%	86.88%
32 Units	2	32, 32	88.97%	86.78%
64 Units	2	64, 64	89.07%	86.86%
MSE Loss	2	16, 16	88.73%	87.69%
Tanh Activation	2	16, 16	88.88%	86.20%
Dropout 0.2	2	16, 16	88.80%	87.11%
Dropout 0.5	2	16, 16	88.80%	88.09%
L2 Regularization	2	16, 16	88.88%	86.59%

Overall, the validation accuracy results were relatively close. The highest validation accuracy was 89.17% from the baseline model. The highest test accuracy was 88.09% from the model using 50% dropout. One of the main findings was that making the neural network larger did not automatically make it better. Adding a third hidden layer or increasing the number of hidden units produced results that were very similar to the baseline. The training results also showed signs of overfitting. In several models, training accuracy continued to increase while validation accuracy stopped improving or began to decrease. This means the model was becoming better at fitting the training data without improving its performance on new data. The dropout method were useful because dropout helps prevent the model from relying too heavily

on individual units during training. The 50% dropout model had the highest test accuracy in these models, even though its validation accuracy was not the highest.

Another important finding was that validation accuracy and test accuracy were not always the same. The model with the highest validation accuracy was not the model with the highest test accuracy. This shows why both validation and test performance are important when evaluating a machine learning model.

The models demonstrated that different neural network settings can produce different results, but larger and more complicated models did not automatically produce better performance. The baseline two hidden layer model had the highest validation accuracy at 89.17%. The model using 50% dropout had the highest test accuracy at 88.09%. The results also showed that the choice of architecture, loss function, activation function, and regularization technique can affect how well a model performs on data it has not seen before.

For this set of tests, the two hidden layer baseline model provides a useful model for comparison because it produced the highest validation accuracy. The 50% dropout model also produced an important result because it achieved the highest test accuracy. The results suggest that model selection should consider both validation and test performance rather than relying on training accuracy alone. Future testing could include additional dropout rates, different learning rates, and different numbers of training epochs to determine whether performance can be improved further.